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THE DYNAMIC MODEL FOR CHILLING EVALUATION IN PEACH BUDS

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Abstract

The dynamic model performance is demonstrated by using various temperature conditions. The model is based on experimentation with small peach plants tested for their response to chilling under strictly controlled conditions. Its effect was verified in Israel and in other countries. It may serve both as a tool for research and as an aid for growers to evaluate the development of dormancy in their tree buds.

Résumé

La performance du modèle dynamique est démontrée en utilisant diverses conditions de température. Le modèle est basé sur une expérimentation avec des petits arbres dont la réponse au froid a été testée sous des conditions contrôlées strictes. Ses effets ont été vérifiés en Israel et dans d'autres pays. Il peut servir à la fois comme outil de recherche et comme aide aux producteurs pour évaluer l'évolution de la dormance des bourgeons de leurs arbres.

1. Introduction

The fact that buds of temperate-zone fruit trees require chilling in order to break properly in spring was first noted by Covile in 1920. Since then, various climatic models were developed to describe the response of temperate fruit tree buds to temperature. A detailed study by Erez *et al.* (1979a,b), Couvillon and Erez (1985) and Erez and Couvillon (1987) set the basis for the development of the Dynamic Model (Fishman *et al.*, 1987a,b; Erez *et al.*, 1988; Erez *et al.*, 1990) which is based on the following findings:

1. Chilling effect shows an optimum curve with maximum at 6-8°C with low effects at 12 and 0°C.

2. Chilling negation by high temperatures is level, duration and cycle dependent. With higher temperatures, longer duration in a daily cycle and with short cycles (like a daily cycle) the chilling negation is stronger. Short exposure to temperatures of 20°C will not be enough to negate chilling. Likewise exposure to high temperatures in a cycle of a few days will not negate exposure to former chilling due to fixation of the chilling effect.

3. Chilling effect can be further enhanced by moderate temperature if it alternates in a daily cycle with low temperatures. Moderate temperatures in the range 13 to 16°C will thus enhance the chilling effect.

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2. Description of the model principle

As a result of work cited above, and based on the finding that chilling reversal by high temperature is limited to short cycles, a detailed model was developed that was termed 'The Dynamic Model'. The model is based on the following elements (Fishman *et al.*, 1987a,b,):

A -The two step system concept: the first one builds an intermediate that is accumulated when exposed to low temperature. The intermediate level depends on:

1. The bell shape curve effect of chilling;
2. The negating of chilling by high temperatures (effect of level; effect of high temperature duration; effect of cycle);
3. The promotive effect of moderate temperatures.

B -The concept of a fixation effect: When a critical level of the intermediate is being reached, a phase transition occurs, the intermediate level drops to 0 and 1 quantum that is termed "Chilling portion" is accumulated. This transfer is automatic at temperatures above 4°C but depends on temperature >4°C

C - The concept of a quantum: When a portion, which size is physiological, is accumulated, it is fixed and conserved.

D - The concept of a threshold level: A critical level of the intermediate has to be reached for chilling to accumulate. As long as this threshold is not reached, no matter how close the level of the intermediate is to it, no chilling accumulation will occur.

The difference among cultivars or species is in the total portions needed for breaking their dormancy not in the model parameters. A partial list of chilling portions required is shown in Table 1.

By analyzing climatic data from various warm locations where temperate fruit trees are grown, it was found that the main cause for poor bud is the increase in day temperatures and not the lack of low night temperatures. So it is not the lack of chilling that is responsible for poor bud break but the negation of chilling by high day-time temperature that changes in a diurnal pattern. The effect of specific sets of temperatures on dormancy release is demonstrated by running the "Stella" simulation computer program with the data. Evaluation of the advantage of evaporative cooling to counteract excessive day temperatures is shown in Fig. 1. A typical warm winter conditions is depicted in the nonsprinkled control for a period of 144 hours. Four chilling portions did accumulate during this period when high day temperature was reduced by evaporative cooling in days 3, 4 and 5 as against only 2 portions in the non sprinkled control.

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Table 1 - Evaluation of number of chilling portions* needed to satisfy the chilling requirements** of various peach and nectarine cultivars.
( ) - for vegetative buds if required different chilling

Cultivar	Chilling portions
Flordaprince	12
Maravilha,	18 (22)
Earligrande	18
Flordaglo	18 (22)
83-4W	18 (25)
Flordagold	26
Rhodes	30
Summerset	40
Redhaven	43
Aprilglo (nect.)	12
Fiestared (nect.)	18
Sunblaze (nect.)	20
Mayglo (nect.)	22
Flavortop (nect.)	36
Fantasia (nect.)	42

* One chilling portion is equivalent to 30 hours at continuous chilling at 6°C
** Requirement was defined as the chilling required to obtain equal or higher bud break than 70% of all buds

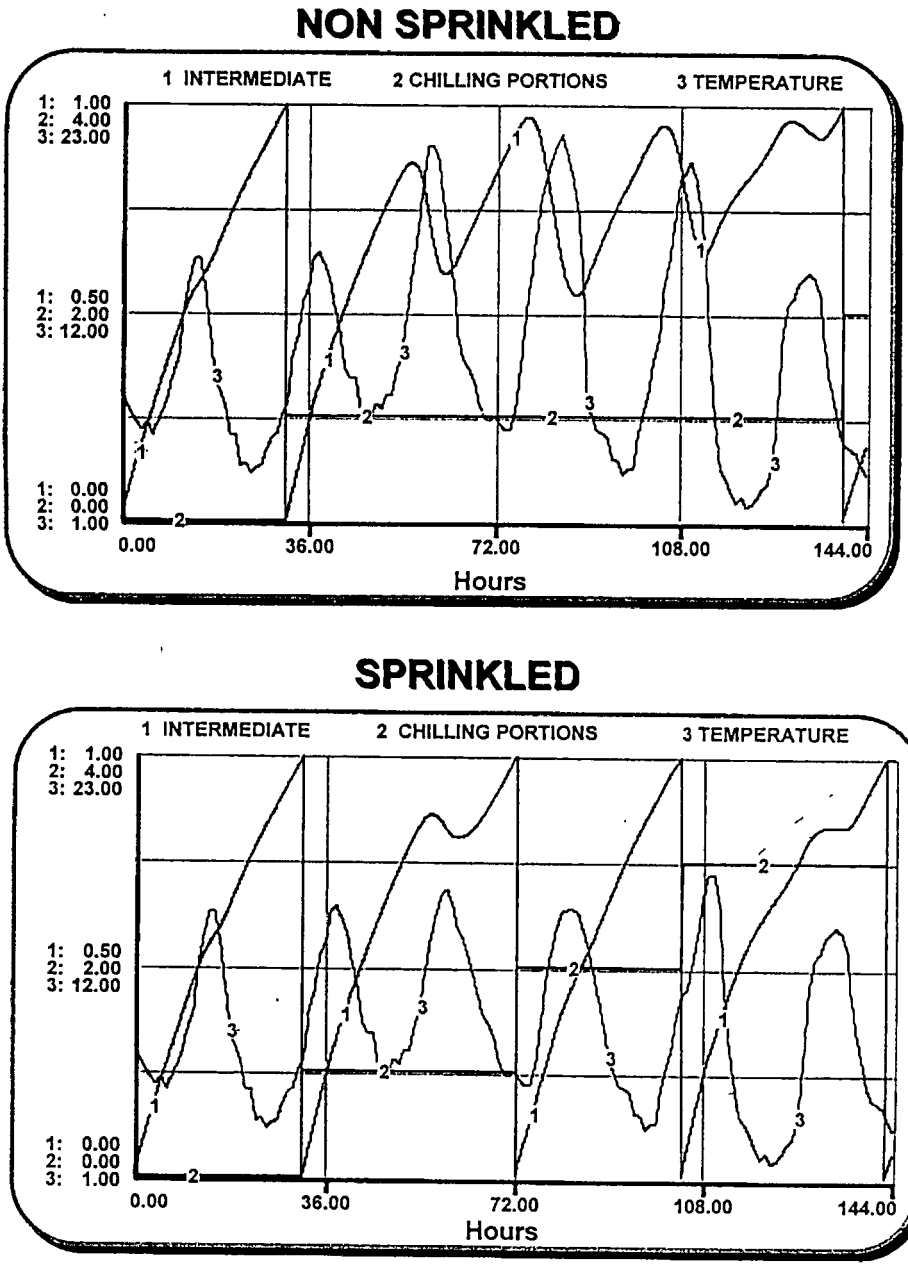


Figure 1 - Temperature pattern versus intermediate level and portions accumulated in nonsprinkled control and sprinkled treatment during 144 hours. The model ran with the same data but with a change in days 3, 4 and 5 when overhead sprinkling reduced high day-time temperatures. 1= intermediate level; 2 = chilling portions and 3 = bud temperature.